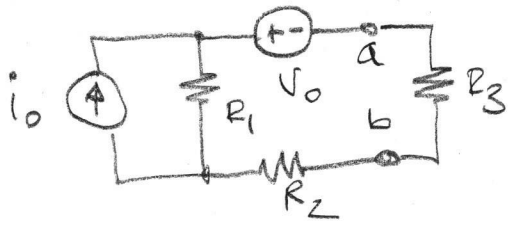
Lösningsförslag

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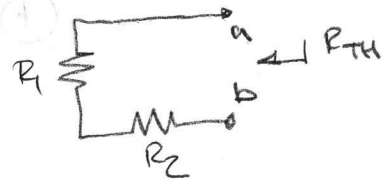
1)



$R_1 = 30\,\Omega$ $V = 10\,V$
$R_2 = 20\,\Omega$
$R_3 = 40\,\Omega$
$i_0 = 0.76\,A$

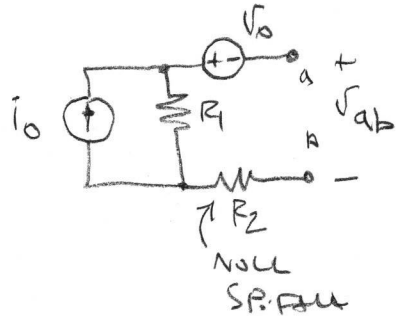
a) THEVENINEKVIVALENT. R_3 BORTKOPPLAT

BESTÄM $R_{TH} = R_{ab}$



$R_{TH} = R_1 + R_2 = \underline{50\,\Omega}$

BESTÄM $V_{ab} = V_{TH}$



OHMS LAG GER

$V_{R_1} = R_1 i_0 = 0.76 \cdot 30 =$

$= \underline{22.8\,V}$

KSPL GER

$-V_{ab} + V_{R_1} - V_0 = 0$

$V_{ab} = V_{R_1} - V_0 = 22.8 - 10 =$

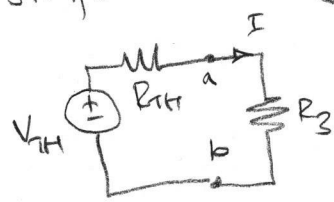
$= \underline{12.8\,V}$

THEVENINKRETSEN BLIR



$\begin{cases} R_{TH} = 50\,\Omega \\ V_{TH} = 12.8\,V \end{cases}$

b) SP. FALL ÖVER R_3



$I = \dfrac{V_{TH}}{R_{TH} + R_3} = \dfrac{12.8}{40+50} = \underline{0.142\,A}$

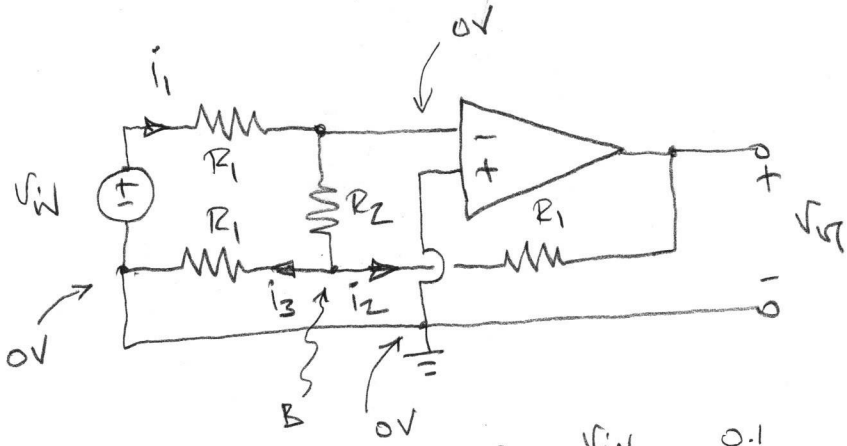
$V_{R_3} = R_3 I = 40 \cdot 0.142 =$

$= \underline{5.68\ \text{VOLT}}$

SVAR: a) $V_{TH} = 12.8\,V$
$R_{TH} = 50\,\Omega$

b) $V_{R_3} = 5.68$ VOLT

2)



$R_1 = 10\,K\Omega$

$R_2 = 100\,K\Omega$

$V_{IN} = 0.1$ VOLT

OHMS LAG GER $i_1 = \frac{V_{IN}}{R_1} = \frac{0.1}{10 \cdot 10^3} = \underline{10 \cdot 10^{-6}}$ A (1)

FÖR NOD B GÄLLER KIRCHOFFS STRÖMLAG

$$i_1 - i_2 - i_3 = 0 \qquad (2)$$

KSPL GER

$$V_{IN} - R_1 i_1 - R_2 i_1 - R_3 i_3 = 0$$

$$i_3 = \frac{1}{R_3}\left(V_{IN} - (R_1 + R_2) i_1\right) =$$

$$= \frac{1}{10 \cdot 10^3}\left(0.1 - 110 \cdot 10^3 \cdot 10 \cdot 10^{-6}\right) = \underline{-0.1 \cdot 10^{-3}}\ A \qquad (3)$$

(3) i (2) GER

$$i_2 = i_1 - i_3 = 10 \cdot 10^{-6} - (-0.1 \cdot 10^{-3}) = \underline{110 \cdot 10^{-6}}\ A$$

KSPL igen

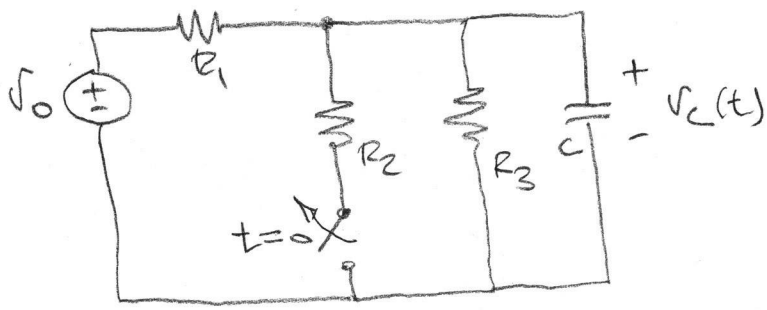
$$V_{UT} + R_1 i_2 - R_1 i_3 = 0$$

$$V_{UT} = R_1 i_3 - R_1 i_2 = 10 \cdot 10^3 (-0.1 \cdot 10^{-3}) -$$

$$- 10 \cdot 10^3 (110 \cdot 10^{-6}) = \underline{-2.1\ V}$$

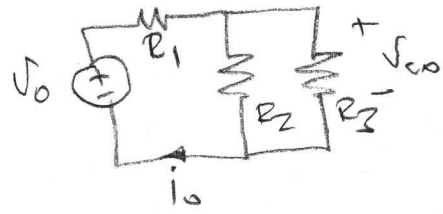
SVAR: $V_{UT} = -2.1$ V

3)



$V_0 = 3V$
$R_1 = 5k\Omega$
$R_2 = 2k\Omega$
$R_3 = 8k\Omega$
$C = 1\mu F$

För $t<0$ gäller



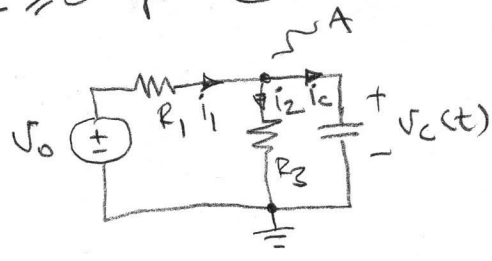
$$i_0 = \frac{V_0}{R_1 + \frac{R_2 R_3}{R_2+R_3}} = \frac{3}{5k + \frac{2k\cdot 8k}{2k+8k}} = 4.54\cdot 10^{-4} A \quad (1)$$

Spänningen över R_3 är ju spänningen över C för $t<0$

dvs
$$V_C(0) = V_{R_3}(t<0) \quad (2)$$

där
$$V_{R_3} = V_0 - R_1 i_0 = 3 - 5\cdot 10^3 \cdot 4.54\cdot 10^{-4} = 0.727 \text{ volt} \quad (3)$$

För $t \geq 0$ gäller



För nod A gäller

$$i_1 - i_2 - i_C = 0 \quad (4)$$

där

$$i_1 = \frac{V_0 - V_A}{R_1}$$
$$i_2 = \frac{V_A}{R_3} \quad (5)$$
$$i_C = C\frac{dV_A}{dt}$$

(5) i (4) ger

$$\frac{V_0 - V_A}{R_1} - \frac{V_A}{R_3} - C\frac{dV_A}{dt} = 0$$

Stuvat om

$$\frac{dV_A}{dt} + \frac{1}{C}\left(\frac{1}{R_1} + \frac{1}{R_3}\right)V_A = \frac{1}{CR_1}V_0 \quad (6)$$

Lösningen till (b) är

$$v_c(t) = Ae^{-\xi t} + B$$

där $\xi = \frac{1}{C}\left(\frac{1}{R_1}+\frac{1}{R_3}\right)$, och B fås ur $B\frac{1}{C}\left(\frac{1}{R_1}+\frac{1}{R_3}\right) = \frac{V_0}{CR_1}$

$$B = \frac{\frac{1}{R_1}}{\frac{1}{R_1}+\frac{1}{R_3}} V_0 =$$

$$= \frac{R_3 V_0}{R_1 + R_3}$$

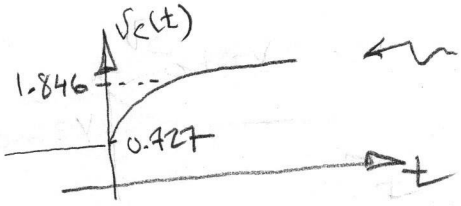
Vi vet att $v_{R_3} = v_c(0) \Rightarrow v_{R_3} = A + B \Rightarrow A = v_{R_3} - B$

Med siffror

$$A = 0.727 - \frac{8K \cdot 3}{5K+8K} = -1.119 \text{ V}$$

Alltså blir spänningen över kondensatorn

$$v_c(t) = -1.119e^{-325t} + 1.846 \text{ volt}$$

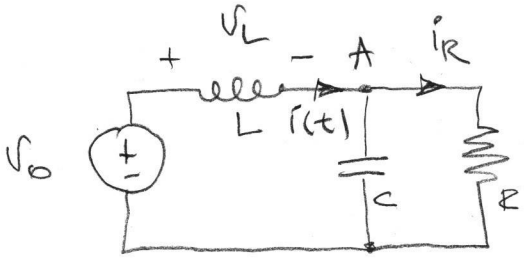


$v_c(\infty) = 1.846$ volt

Svar: a) $v_c(0^-) = 0.727$ volt

b) $v_c(t) = -1.119e^{-325t} + 1.846$ V

4)



$L = 100\,mH$

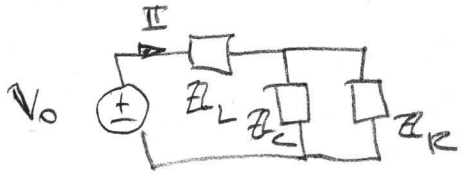
$C = 2.5\,\mu F$

$R = 200\,\Omega$

$V_0 = 10\cos(1000t + 30°)$ ✓

BÖRJAR MED ATT BESTÄMMA STRÖMMEN $i(t)$

HOPPAR ÖVER TILL KOMPLEX REPRESENTATION



TOTALA IMPEDANSEN BLIR

$$Z_{TOT} = Z_L + \frac{Z_C Z_R}{Z_C + Z_R}$$

MEN

$$Z_L = j\omega L \,,\quad Z_C = \frac{1}{j\omega C} \,,\quad Z_R = R$$

$$Z_{TOT} = j\omega L + \frac{\frac{1}{j\omega C} R}{R + \frac{1}{j\omega C}} = j\omega L + \frac{R}{1 + j\omega CR}$$

SIFFROR

$$Z_{TOT} = j1000 \cdot 100 \cdot 10^{-3} + \frac{200}{1 + j1000 \cdot 2.5 \cdot 10^{-6} \cdot 200} =$$

$$= j100 + \frac{200}{1 + j0.5} = j100 + \frac{200}{1.12\angle 26.56°} =$$

$$= 100\angle 90° + 178.57\angle -26.56 =$$

$$= 159.72 + j20.16 = 160.99\angle 7.19°\ \Omega$$

STRÖMMEN BLIR

$$II = \frac{V_0}{Z_{TOT}} = \frac{10\angle 30°}{160.99\angle 7.19°} = 0.062\angle 22.81°\ A$$

SPÄNNINGEN ÖVER SPOLEN BLIR

$$V_L = Z_L II = j1000 \cdot 100 \cdot 10^{-3} \cdot 0.062\angle 22.81° =$$

$$= 6.2\angle 112.81°\ VOLT$$

STRÖMMEN GENOM MOTSTÅNDET BERÄKNAS AV NODSP. I A.

DVS

$$\mathbb{I}_R = \frac{\mathbb{V}_A}{R}$$

DÄR

$$\mathbb{V}_A = \mathbb{V}_0 - \mathbb{V}_L = 10\angle 30^\circ - 6.2\angle 112.81^\circ =$$

$$= 11.064 - 0.715j = 11.09\angle -3.70^\circ \text{ VOLT}$$

TILLBAKA TILL TIDSPLANET, TAG REALDELEN AV SVAREN

$$v_L(t) = \mathrm{Re}\{\mathbb{V}_L\} = 6.2\cos(1000t + 112.81^\circ) \text{ V}$$

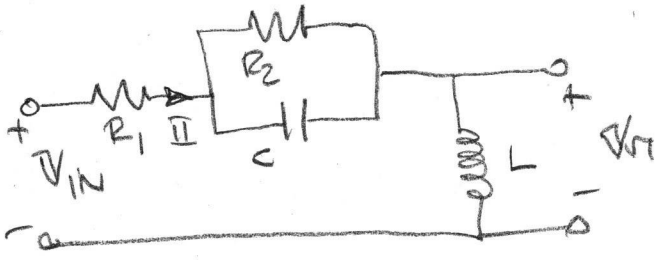
och

$$i_R(t) = \mathrm{Re}\left\{\frac{\mathbb{V}_A}{R}\right\} = 0.0554\cos(1000t - 3.70^\circ) \text{ A}$$

SVAR: $v_L(t) = 6.2\cos(1000t + 112.81^\circ)$ V

$i_R(t) = 55.4\cos(1000t - 3.70^\circ)$ mA

5)



KSPL ger

$$V_{IN} - R_1 II - Z_p II - j\omega L II = 0 \qquad (1)$$

$$V_{UT} - j\omega L II = 0 \Rightarrow II = \frac{V_{UT}}{j\omega L} \qquad (2)$$

(2) i (1) ger

$$V_{IN} = (R_1 + Z_p + j\omega L)\frac{1}{j\omega L} V_{UT}$$

$$\frac{V_{UT}}{V_{IN}} = \frac{j\omega L}{(R_1 + Z_p + j\omega L)} \qquad (3)$$

P-kopplingen

$$Z_p = \frac{\frac{1}{j\omega C} R_2}{R_2 + \frac{1}{j\omega C}} = \frac{R_2}{1 + j\omega C R_2} \qquad (4)$$

(4) i (3) ger

$$H(\omega) = \frac{V_{UT}}{V_{IN}} = \frac{j\omega L}{R_1 + \frac{R_2}{1+j\omega C R_2} + j\omega L} =$$

$$= \frac{j\omega L (1 + j\omega C R_2)}{R_1 + R_2 + j\omega C R_1 R_2 + j\omega L - \omega^2 L C R_2} =$$

$$= \frac{-\omega^2 L C R_2 + j\omega L}{R_1 + R_2 - \omega^2 L C R_2 + j(\omega C R_1 R_2 + \omega L)} \qquad (5)$$

u.

FÖRKORTNING MED KOMPLEXKONJUGATET

$$H(\omega) = \frac{(-\omega^2 LCR_2 + j\omega L)((R_1+R_2-\omega^2 LCR_2 - j(\omega CR_1R_2+\omega L))}{\underbrace{(R_1+R_2-\omega^2 LCR_2)^2 + (\omega CR_1R_2+\omega L)^2}_{=N}} \quad (6)$$

MULTIPLICERA UT:

$$-\omega^2 LCR_2(R_1+R_2-\omega^2 LCR_2) + j\omega L(R_1+R_2-\omega^2 LCR_2) +$$
$$+j\omega^2 LCR_2(\omega CR_1R_2+\omega L) + \omega L(\omega CR_1R_2+\omega L) \quad (6.5)$$

DÅ FÅS

$$Re\{H(\omega)\} = \frac{1}{N}(-\omega^2 LCR_2(R_1+R_2-\omega^2 LCR_2) +$$
$$+\omega L(\omega CR_1R_2+\omega L)) \quad (6.6)$$

$$IM\{H(\omega)\} = \frac{1}{N}(\omega L(R_1+R_2-\omega^2 LCR_2) +$$
$$+\omega^2 LCR_2(\omega CR_1R_2+\omega L)) \quad (6.7)$$

MED DET SÅ KAN FASFUNKTIONEN LÄTT BERÄKNAS

$$\Phi(\omega) = ATAN\left(\frac{IM\{H(\omega)\}}{Re\{H(\omega)\}}\right) =$$

$$= ATAN\left(\frac{\omega L(R_1+R_2-\omega^2 LCR_2)+\omega^2 LCR_2(\omega CR_1R_2+\omega L)}{-\omega^2 LCR_2(R_1+R_2-\omega^2 LCR_2)+\omega L(\omega CR_1R_2+\omega L)}\right) =$$

$$= ATAN\left(\frac{R_1+R_2-\omega^2 LCR_2+\omega^2 C^2 R_1R_2^2+\omega^2 LCR_2}{\omega CR_1R_2+\omega L-\omega CR_2(R_1+R_2-\omega^2 LCR_2)}\right) =$$

$$= ATAN\left(\frac{R_1+R_2+\omega^2 C^2 R_1R_2^2}{\omega CR_1R_2+\omega L-\omega CR_2(R_1+R_2-\omega^2 LCR_2)}\right) \quad (7)$$

SVAR:

$$\begin{cases} \mathbb{H}(\omega) = \dfrac{-\omega^2 LCR_2 + j\omega L}{R_1 + R_2 - \omega^2 LCR_2 + j(\omega CR_1R_2 + \omega L)} & (8) \\ \varphi(\omega) = \text{ATAN}\left(\dfrac{R_1 + R_2 + \omega^2 C^2 R_1 R_2^2}{\omega CR_1R_2 + \omega L - \omega CR_2(R_1 + R_2 - \omega^2 LCR_2)} \right) & (9) \end{cases}$$

för PLOTTAR PÅ $|\mathbb{H}(\omega)|$ och $\varphi(\omega)$ SE NÄSTA SIDA. ANVÄNDA VÄRDEN

$R_1 = 10\,\Omega$, $L = 10\,\text{MH}$
$R_2 = 1\,\text{K}\Omega$, $C = 0.1\,\text{NF}$

b) REALDELEN AV $\mathbb{H}(\omega)$ ÄR, FRÅN (6.6)

$$\text{Re}\{\mathbb{H}(\omega)\} = \left[-\omega^2 LCR_2(R_1 + R_2 - \omega^2 LCR_2) + \omega^2 L(CR_2R_1 + L)\right]\frac{1}{N}$$

MEN OM $\mathbb{H}(\omega)$ SKALL VARA IMAGINÄRT SÅ MÅSTE $[\ldots] = 0$

DVS

$$\omega_0^2 CR_2(R_1 + R_2 - \omega_0^2 LCR_2) = \omega_0^2 L(CR_2R_1 + L)$$

$$CR_2^2 - \omega_0^2 LC^2R_2^2 = L$$

$$\omega_0^2 = \frac{CR_2^2 - L}{LC^2R_2^2} \qquad (10)$$

OM $\underline{CR_2^2 - L > 1}$ SÅ KAN $\mathbb{H}(\omega)$ BLI IMAGINÄR.

OCH $\omega_0 = \sqrt{\dfrac{CR_2^2 - L}{LC^2R_2^2}} = \dfrac{1}{CR_2}\sqrt{\dfrac{CR_2^2}{L} - 1}$

FASFUNKTIONEN BLIR

$$\varphi(\omega_0) = \text{ATAN}\left(\frac{\text{IM}(\mathbb{H}(\omega_0))}{0}\right) = \frac{\pi}{2} \qquad (11)$$

ÅTERSTÅR $|\mathbb{H}(\omega_0)|$ ... NU KOMMER DET KRÅNGLIGA, HÄR GÄLLER DET ATT RÄKNA RÄTT!

▷ forts.

MED (5) FÅS

$$|H(\omega_0)|^2 = \frac{(\omega_0^2 LCR_2)^2 + (\omega_0 L)^2}{(R_1+R_2-\omega_0^2 LCR_2)^2 + (\omega_0 CR_1R_2+\omega_0 L)^2}$$

TAR TÄLJAREN OCH NÄMNAREN VAR FÖR SIG. MED (10) FÅS

$$T = \left(\frac{CR_2^2-L}{LC^2R_2^2}\right)^2 L^2C^2R_2^2 + \frac{CR_2^2-L}{LC^2R_2^2}L^2 =$$

$$= \frac{CR_2^2-L}{LC^2R_2^2}\left[\frac{CR_2^2-L}{LC^2R_2^2}L^2C^2R_2^2 + L^2\right] =$$

$$= \frac{CR_2^2-L}{LC^2R_2^2}\left[CR_2^2L - L^2 + L^2\right] = \frac{CR_2^2-L}{LC^2R_2^2}CR_2^2L =$$

$$= \frac{CR_2^2-L}{C} = R_2^2 - \frac{L}{C} \qquad (12)$$

NÄMNAREN

$$N = \left(R_1+R_2 - \frac{CR_2^2-L}{LC^2R_2^2}LCR_2\right)^2 + \frac{CR_2^2-L}{LC^2R_2^2}(CR_1R_2+L)^2 =$$

$$= \left(R_1+R_2 - R_2 + \frac{L}{CR_2}\right)^2 + \frac{CR_2^2-L}{LC^2R_2^2}(CR_1R_2+L)^2 =$$

$$= \frac{1}{C^2R_2^2}(CR_1R_2+L)^2 + \frac{CR_2^2-L}{LC^2R_2^2}(CR_1R_2+L)^2 =$$

$$= (CR_1R_2+L)^2\left(\frac{1}{C^2R_2^2} + \frac{CR_2^2-L}{LC^2R_2^2}\right) =$$

$$= (CR_1R_2+L)^2\left(\frac{L}{CR_2^2} + \frac{CR_2^2}{L} - \frac{L}{CR_2^2}\right)\frac{1}{C^2R_2^2} = (CR_1R_2+L)^2\frac{1}{CL} \qquad (13)$$

(13) & (12) ger

$$|H(\omega_0)|^2 = \frac{CL\left(R_2^2 - \frac{L}{C}\right)}{(CR_1R_2+L)^2}$$

ALTSÅ $$|H(\omega_0)| = \frac{1}{CR_1R_2 + L}\sqrt{CL\left(R_2^2 - \frac{L}{C}\right)}$$

$\lim_{\omega \to \infty} |H(\omega)| = 1$ (VR (8))

ALTSÅ $|H(\omega)| > 1$ om $\dfrac{\sqrt{CL\left(R_2^2 - \frac{L}{C}\right)}}{CR_1R_2 + L} > 1$.

SLUT!

FOTNOT: I samband med att uppgift 5b BARA ALGEBRAISKT SVÅR är uträkningen uppgift 5b. Uppgift 5a går också på fler än 5p.

SVAR : a) $$|H(\omega)| = \frac{-\omega^2 LCR_2 + j\omega L}{R_1 + R_2 - \omega^2 LCR_2 + j(\omega CR_1R_2 + \omega L)}$$

$$\varphi(\omega) = \text{ATAN}\left(\frac{R_1 + R_2 + \omega^2 C^2 R_1 R_2^2}{\omega CR_1R_2 + \omega L - \omega CR_2(R_1 + R_2 - \omega^2 LCR_2)} \right)$$

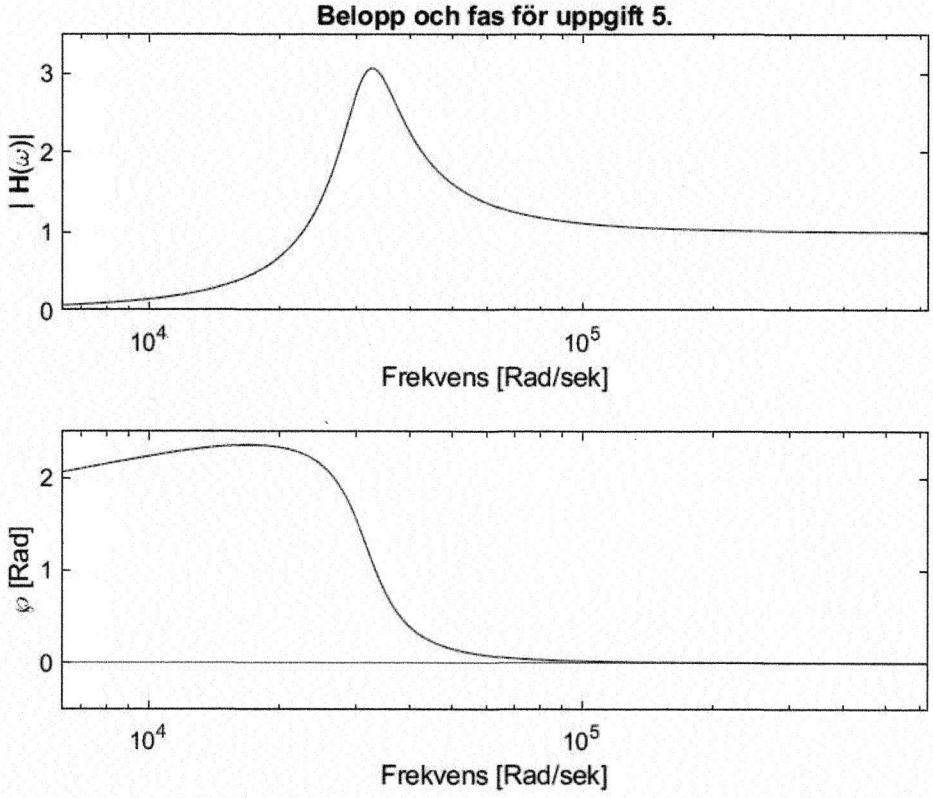
b) $$\omega_0^2 = \frac{CR_2^2 - L}{LC^2R_2^2}$$

$CR_2^2 - L > 1$ DÅ BLIR

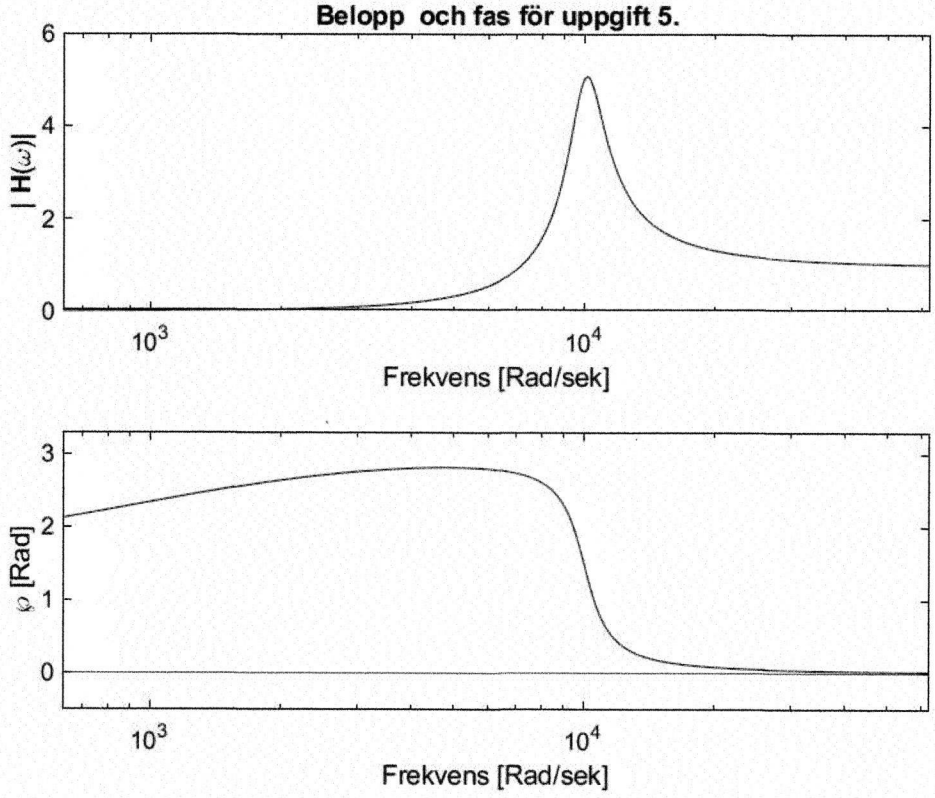
$$\begin{cases} \varphi(\omega_0) = \frac{\pi}{2} \\ |H(\omega_0)| = \left[\frac{CL\left(R_2^2 - \frac{L}{C}\right)}{(CR_1R_2 + L)^2} \right]^{1/2} \end{cases}$$

PLOT FÖR VÄRDEN SOM UPPFYLLER VILLKOREN

SE SIDA 14.



$$\begin{cases} R_1 = 10\,\Omega, \; R_2 = 1\,K\Omega \\ L = 10\,MH, \; C = 0.1\,\mu F \end{cases}$$



$R_1 = 10\ \Omega$, $R_2 = 1\text{K}\Omega$

$L = 10\ \text{mH}$, $C = 1\text{NF}$